

Question	Marking guidance	Mark	Comments
03.1	Uses $T = I\alpha$ ✓ applied torque = 440×0.090 OR 39.6 (N m) ✓ 52 or 51.6 rad (s^{-2}) ✓	3	Either correct substitution or rearrangement Award MP2 if (net torque =) 35.6 seen
03.2	31 (rad s^{-1}) ecf ✓	1	Ecf from 03.1 ($0.6 \times$ answer to 03.1)
03.3	Use of $w = \frac{v}{r}$ leading to 117 to at least 3 sf ✓	1	Not 116 (rounding error)
03.4	Use of $E_k = \frac{1}{2}mv^2$ or $E_k = \frac{1}{2}I\omega^2$ ✓ Correct kinetic energy for bike (1.14×10^5) or for one wheel (4968) or for both wheels (9936) ✓ Finds rotational E_k as a percentage of translational E_k or total E_k ✓ Total kinetic energy = 1.23×10^5 (J) ✓ $8.0(3)\%$ based on $\omega=120$ OR 7.6% based on $\omega=117$ ✓	4	Accept 4696 (1 wheel) or 9392 (2 wheels) based on $\omega=117$ for MP2 Condone use of only 1 wheel for MP3 Accept reasonable rounding error in final answer
03.5	reduces moment of inertia or mass ✓ Linked argument to show that acceleration and / or deceleration will be greater (for same torque) ✓ Uses $t = I\alpha$ or uses $F = ma$ to link to acceleration ✓	3	Or less torque needed to produce same acceleration (MP2)

Question	Marking guidance	Mark	Comments
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	32	D
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Question	Marking guidance	Additional comments/guidelines	Mark
04.1	Uses $T = I \alpha$ ✓ Uses $\omega = \alpha t$ ✓ Both of the above used 10.6 (rad s ⁻¹) to at least 3 sf ✓	Look for (α) 55/37 or 1.49	3
04.2	390 or 392 ✓ kg m ² s ⁻¹ ✓	Use of $\omega = 11$ gives 410 or 407 Accept kg m ² rad s ⁻¹	2
04.3	Angular speed will increase ✓ Moment of inertia (of system) decreases ✓ Reference to principle of conservation of angular momentum ✓	Do not credit answers that lead to angular speed stays the same/decreases	3
Total			8

Question	Marking guidance	Additional comments/guidelines	Mark
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	31	B
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	32	B
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	33	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Use of angular deceleration = change in angular velocity / time AND Use of $\tau = I\alpha$ OR Use of $\tau t = I\Delta\omega$ ✓ $t = 2.5$ (s) ✓	Condone one error in manipulation or substitution for MP1	2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Finds the difference in moment of inertia = $8.1 \times 10^{-5} - 3.5 \times 10^{-5}$ ✓ Use of $I = 3mr^2$ ✓ ($m =$) 7.2×10^{-3} kg ✓	7.2, 7.25, 7.246 not 7.3	3	3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $\tau = I\alpha$ ✓ Use of $\alpha = \frac{\Delta\omega}{t}$ ✓ Adds resultant torque and frictional torque OR use of $\tau = Fr$ ✓ ($F =$) 0.044 (N) ✓	alternative for MP1 and MP2: use of $\tau = \frac{\Delta J}{t}$ ✓✓ For MP3, full substitution and manipulation needed	4	4 × AO3

Total			3	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Evidence of determining an angle of rotation between 146° and 151° ✓ Multiplies by $\frac{\pi}{180}$ or $\frac{2\pi}{360}$ to give a value that rounds to 2.6 ✓	Alternative method: Measures arc length (~90 mm) and radius (35 mm) ✓ Uses $\theta = \frac{s}{r}$ to give a value that rounds to 2.6 ✓	2	AO3 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	1.2 (s) ✓	$\theta = \frac{1}{2}\alpha t^2$	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $\omega^2 = 2\alpha\theta$ OR $\theta = \frac{1}{2}\alpha t^2$ with $\omega = \alpha t$ ✓ 6.7 (rad s ⁻¹) ✓	Expect $\theta = 2\pi$. Condone $\theta = \pi$ for MP1.	2	AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Moment of inertia of X is greater (than Y) ✓ Torque is the same for X and Y ✓ Refers to $T = I\alpha$ to state that (angular) acceleration of X is smaller (than Y) ✓	Comment about difference in mass negates MP1. Accept reverse arguments for MP1 and MP3.	3	AO3 AO2 AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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20	A	$\frac{2mL^2}{3}$	AO2
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21	A	8 clockwise	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Evidence of use of Mr^2 ✓ $r = 6.8 \times 10^{-2} \text{ (m)}$ ✓	Can be by re-arrangement or substitution Condone use of one mass in MP1 Calculator value is $6.770032 \times 10^{-2} \text{ (m)}$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Evidence of use of a rotational KE equation ✓ $7.3(2) \text{ (rad s}^{-1}\text{)}$ ✓	Calculator value is 7.3236851	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Idea that r decreases so I decreases ✓ Uses $T = I\alpha$ to make conclusion about change in α consistent with their change in I from MP1. ✓	Both marks can be given for a full algebraic approach. Expect ' α increases' for their conclusion.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Idea that some energy is transferred from GPE of the spheres (as they fall). ✓ Idea that work done (by frictional torque) = change in KE / 59 mJ + change in GPE ✓	In MP1 allow idea that the GPE (of the spheres) decreases.	2	AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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21	D	2.6 rad s^{-1}	AO3
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22	A	8	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Use of maximum power available $= \frac{1}{2}\pi r^2 \rho v^3$ $= \frac{1}{2}\pi \left(\frac{135}{2}\right)^2 \times 1.2 \times 11.4^3 \checkmark$ 13 MW $\checkmark$	Allow MP1 with POT errors or if diameter is used (expect 50.8 MW if diameter is used) 12.7 MW	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Idea that not all of the moving air interacts with the blades $\checkmark$ Idea that the wind requires speed/momentum/kinetic energy to leave rear of turbine (so cannot transfer 100%) $\checkmark$	Condone rotor for blades. Condone appropriate part of turbine blocking some of the wind in MP1. Accept reference to the Betz limit (59.3%) for MP2 Do not accept ideas of objects blocking wind in front of turbine. Ignore references to the angle of the turbine.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Uses $f = \frac{\omega}{2\pi}$ to get 13.5 (revolutions min^{-1}) $\checkmark$	Allow 13, reject 14 13.46450819 Condone 13.47	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Uses power = torque $\times$ angular velocity = $4.0 \times 10^6 \times 1.41$ ✓ Uses energy = power $\times$ time = $5.6 \times 10^6 \times 5$ to give 28(.2) (MJ) ✓	Look for 5.6 (MW) but condone use of wrong value for torque for MP1	2	1 $\times$ AO1 1 $\times$ AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	Uses $T = I\alpha$ with T between 0.4×10^6 and 4.4×10^6 ✓ Uses resultant torque = 3.6×10^6 (Nm) to give 0.051 (rad s ⁻²) ✓		2	1 $\times$ AO1 1 $\times$ AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.6	Use of $E_k = \frac{1}{2} I \omega^2$ ✓ 7.1×10^7 (J) ✓	Expect $\frac{1}{2} \times 7.1 \times 10^7 \times 1.41^2$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.7	MAX 3 ✓✓✓ - Angular acceleration reduces with time (for the first 55 s) ...because the (net) torque reduces with time AND reference to $T = I\alpha$ - Idea that opposing torque from the generator increases with time (as the power output of the generator increases) - Angular acceleration is zero from $t = 50$ s to $t = 55$ s since the resistive torque is equal and opposite to the torque applied by the wind (to the rotor)	Accept explanation for changing torque from generator in terms of increasing back emf Do not allow references to torque due to air resistance. Ignore references to forces increasing or decreasing, only credit answers which refer to torques for bullet points 2 and 3. Reject answers which refer to balanced forces for bullet point 4. Accept resultant torque is 0 if resistive torque and torque due to the wind are identified. Condone thrust torque or driving torque for torque applied by the wind.	3	AO3
Total			14	

Question	Answers	Additional comments/Guidelines	Mark	AO
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18	A	P, Q, R	AO3
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19	B	20	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Any two from: ✓✓ • converts 481 revolutions to radian • calculates their work done by frictional torque OR deceleration a of wheel • equates their work done to change in kinetic energy using kinetic energy $= \frac{1}{2} I \omega^2$ OR uses $T = I a$ 0.049 (kg m²) ✓		3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	calculates weight of P ✓ determines angular displacement in radians ✓ uses $T \theta = mg \Delta h$ ✓ $2.6(17) \times 10^{-3}$ (N m) ✓	Tolerate 1 sf answer	4	1 × AO3 3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Any four from: ✓✓✓✓ • take repeat readings and reject anomalous results • (several repeats and) find half-range and use for uncertainty in angle or height • how to use the uncertainty to determine the percentage uncertainty • combine percentage uncertainties in h and θ • use percentage uncertainty to determine absolute uncertainty		4	3 × AO3 1 × AO4
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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15	D	84 kg m^2	AO3
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21	A	0.50 rad s^{-2} clockwise	AO2
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Total marks = 15

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Evidence of attempt at suitable graph indicated by quantities on axes ✓ Suitable scales ✓ Suitable labels of graph axis quantities, with relevant units ✓ MP1 and MP2 and points give a straight line ✓ Analysis to show equation is consistent with the graph ✓	eg R against $A^{\frac{1}{3}}$ OR R^3 against A OR $\log(R)$ against $\log(A)$ For MP2 assume missing origin label is "0". For MP3 do not allow $\ln(R)$ for $\ln(R/fm)$ For MP4 non-log graph should be shown to go through the (0,0) either on the graph or by analysis Eg for MP5 Comparison with $y = mx + c$ or justification in terms of (direct) proportionality OR, if log-log graph used, gradient = 3 or 0.33	5	$2 \times \text{AO4}$ $1 \times \text{AO3}$ $2 \times \text{AO2}$

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	<u>Uses graph</u> to determine correct value of R or R_0 (≈ 1.06 fm) OR Uses their R_0 or R reading to determine their $A^{\frac{1}{3}}$ ✓ 102 based on correct method ✓	Allow alternative MP1 for correct use of R^3 vs A graph or log-log graph. Allow 90 to 120	2	$1 \times \text{AO3}$ $1 \times \text{AO2}$

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Any two from: ✓✓ - evidence of determination of charge for either alpha or sodium - correct conversion of femto to metres - evidence of use potential energy equation 2.0×10^{-13} J based on correct working ✓	Do not penalise POT error in BP1 3.2×10^{-19} or 1.76×10^{-18} In BP1 if no value used for e, allow the mark if 2e AND 11e seen For BP3 do not credit use of coulomb force $\times r$.	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Idea that answer depends on the alpha particle kinetic energy ✓	Allow idea that alpha particle approach only depends on protons Allow idea that alpha particles lose all their KE before reaching the nucleus	1	AO1

Total			11	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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